

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

I _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A
Scenario-based

1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

- A. Explain why this issue occurs.
- B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

- A. Identify the most suitable algorithm for this scenario.
- B. Justify your choice based on security strength and efficiency.

3. Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

- A. Explain how the key exchange works in this scenario.
- B. Identify a possible security threat and suggest a solution to mitigate it.

4. Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

- A. Describe how RSA can be used for encryption and decryption in this scenario.
- B. Explain how key distribution and management play a role in maintaining security.

5. Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

- A. Analyze the constraints of IoT devices.
- B. Recommend a suitable cryptographic approach and justify your answer.

RUBRICS FOR CALCULATION:

Q. No	Understanding of Concepts (25)	Experimental Design (30)	Use of Tools (20)	Analysis of Results (15)	Documentation (10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

CO2 AT2 CRYPTOGRAPHY:

1. Secure Data Transmission using Block Cipher

A. Explain why this issue occurs.

- ECB (Electronic Codebook) encrypts each block independently.
- Identical plaintext blocks produce identical ciphertext blocks.
- This creates visible patterns, making data easier to analyze.

B. Suggest a more secure block cipher mode and justify.

- **Use CBC (Cipher Block Chaining) or GCM mode.**
 - Each block depends on the previous encrypted block (or uses a unique nonce in GCM).
 - This removes patterns and provides better confidentiality.
 - **GCM** also provides authentication and integrity.
-

2. Choosing an Encryption Algorithm for Banking System

A. Identify the most suitable algorithm.

- **Blowfish**

B. Justify your choice.

- Blowfish supports keys up to **448 bits**.
 - More secure than **DES (56-bit)**.
 - Faster and more efficient than **3DES**.
 - Suitable for banking applications requiring high security and moderate performance.
-

3. Key Exchange in an Unsecured Network

A. Explain how Diffie–Hellman key exchange works.

1. Both users agree on a public prime number (**P**) and generator (**G**).
2. Each user selects a private key.
3. They exchange public keys.
4. Each calculates the same shared secret using their private key and the other user's public key.
5. The shared secret is used as the encryption key.

B. Security threat and solution.

Threat: Man-in-the-Middle (MITM) attack.

Solution: Use **authenticated Diffie–Hellman** with **digital certificates or RSA-based authentication** to verify identities.

4. Public Key Cryptography for Email Security

A. How RSA is used for encryption and decryption.

- Sender encrypts the email using the receiver's **public key**.
- Receiver decrypts it using their **private key**.
- This ensures only the intended receiver can read the email.

B. Role of key distribution and management.

- Public keys must be distributed securely.
 - Private keys must remain secret.
 - Digital certificates and a Public Key Infrastructure (PKI) help verify key authenticity and prevent impersonation.
-

5. Cryptography for IoT Devices

A. Analyse the constraints of IoT devices.

- Limited CPU processing power.

- Limited memory and storage.
- Low battery power.
- Limited network bandwidth.

B. Recommend a suitable cryptographic approach.

- **Elliptic Curve Cryptography (ECC)**

Justification:

- Provides strong security with smaller key sizes.
- Faster than RSA on low-power devices.
- Uses less memory, processing power, and battery.
- Ideal for IoT environments.